WORKED SOLUTION

STEP II 2002, Question 9

Mechanics • Projectile Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

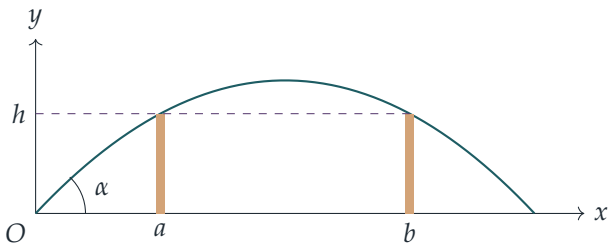
step.maths.org/questions/02-s2-q9

The key idea

Write down the equation of the path, y as a function of x. "Just passes over both walls" says the path goes through the two wall tops, so the equation $y = h$ has roots exactly $x = a$ and $x = b$. That is a quadratic with known roots, and comparing it with its sum and product of roots gives everything at once. The second part is then a matter of differentiating the two formulae with respect to h.

Solution

We model the particle as a point moving under gravity alone (no air resistance), and take O as origin with x horizontal and y vertically up. We assume, as the picture intends, $0 < a < b$, $h > 0$ and $0 < \alpha < \frac{\pi}{2}$.



Finding $\tan\alpha$ and V

With initial velocity components $V\cos\alpha$ and $V\sin\alpha$, after time t

$$x = Vt\cos\alpha, \qquad y = Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(V\cos\alpha)$ gives the path

$$y = x\tan\alpha - \frac{gx^2}{2V^2\cos^2\alpha} = x\tan\alpha - kx^2, \qquad \text{where } k = \frac{g\sec^2\alpha}{2V^2} = \frac{g(1+\tan^2\alpha)}{2V^2} > 0.$$

The particle just clears each wall, so the path passes through the tops (a, h) and (b, h). Hence a and b are roots of

$$x \tan\alpha - kx^2 = h, \qquad \text{that is,} \qquad kx^2 - x\tan\alpha + h = 0.$$

This is a quadratic in x (as $k \neq 0$), so it has at most two roots, and therefore its roots are exactly a and b. Comparing with the sum and product of the roots,

$$a + b = \frac{\tan\alpha}{k}, \qquad ab = \frac{h}{k}.$$

The second gives $k = h/(ab)$, and then the first gives

$$\boxed{\tan\alpha = \frac{h(a+b)}{ab}.}$$

Finally, from the definition of k,

$$\begin{aligned}\frac{2V^2}{g} = \frac{1+\tan^2\alpha}{k} &= \frac{ab}{h}\left(1 + \frac{h^2(a+b)^2}{a^2b^2}\right) \\ &= \frac{ab}{h} + \frac{(a+b)^2 h}{ab},\end{aligned}$$

as required.

(It is worth noticing that with these values the path is $y = \dfrac{h}{ab}\,x(a+b-x)$, which visibly equals h at $x = a$ and $x = b$.)

Raising the walls

The second particle also just clears both walls, now of height $h + \delta h$, at the same distances a and b. So the formulae above hold with h, α, V replaced by $h + \delta h$, $\alpha + \delta\alpha$, $V + \delta V$:

$$\tan\alpha = \frac{a+b}{ab}\,h, \qquad \frac{2V^2}{g} = \frac{ab}{h} + \frac{(a+b)^2}{ab}\,h.$$

Each is a function of h alone (with a, b fixed), so for small changes we may use $\delta(\text{quantity}) \approx \dfrac{\mathrm{d}(\text{quantity})}{\mathrm{d}h}\,\delta h$.

The angle. Differentiating $\tan\alpha = \frac{a+b}{ab}h$ with respect to h gives $\sec^2\alpha\,\dfrac{\mathrm{d}\alpha}{\mathrm{d}h} = \dfrac{a+b}{ab}$, so

$$\sec^2\alpha\,\delta\alpha \approx \frac{a+b}{ab}\,\delta h.$$

Since $\sec^2\alpha > 0$, $a, b > 0$ and $\delta h > 0$, the right-hand side is positive, so $\delta\alpha > 0$: the particle must be thrown more steeply.

Ask yourself The "$\approx$" is only first-order. Can we really conclude the sign of $\delta\alpha$ from it?

Yes, and in fact without approximating at all. The formula for $\tan\alpha$ is linear in h, so exactly $\tan(\alpha+\delta\alpha)-\tan\alpha = \frac{a+b}{ab}\,\delta h > 0$. As tan is strictly increasing on $(0, \frac{\pi}{2})$, this forces $\alpha+\delta\alpha > \alpha$, so $\delta\alpha > 0$.

The speed. Differentiating $\dfrac{2V^2}{g}$ with respect to h,

$$\frac{4V}{g}\frac{\mathrm{d}V}{\mathrm{d}h} = -\frac{ab}{h^2} + \frac{(a+b)^2}{ab} = \frac{(a+b)^2}{abh^2}\left(h^2 - \frac{a^2b^2}{(a+b)^2}\right),$$

so

$$\delta V \approx \frac{g(a+b)^2}{4Vabh^2}\left(h - \frac{ab}{a+b}\right)\left(h + \frac{ab}{a+b}\right)\delta h.$$

Every factor except $h - \frac{ab}{a+b}$ is positive (as $V, a, b, h, \delta h > 0$). Hence δV has the sign of $h - \frac{ab}{a+b}$: it is positive if $h > ab/(a+b)$ and negative if $h < ab/(a+b)$, as required. (Since this derivative is non-zero away from $h = ab/(a+b)$, the first-order term really does fix the sign for small enough δh.)

Interpretation. Comparing with $\tan\alpha = h(a+b)/(ab)$, the condition $h > ab/(a+b)$ is exactly $\tan\alpha > 1$, that is $\alpha > 45°$. As a function of h, $2V^2/g$ has the form $p/h + qh$, which is smallest when $h = \sqrt{p/q} = ab/(a+b)$, where $\alpha = 45°$ and $2V^2/g = 2(a+b)$. So: if the throw is already steeper than 45°, raising the walls needs more speed; if it is shallower, raising the walls (and steepening the throw towards 45°) actually needs less.

Checks

- Take $a = 3\,\mathrm{m}$, $b = 7\,\mathrm{m}$, $h = 2.5\,\mathrm{m}$. Then $\tan\alpha = 25/21$ and $2V^2/g = 8.4 + 11.90\ldots = 20.30\ldots\,\mathrm{m}$. Stepping the motion numerically with these values, the particle is at height 2.5 m (to 10^{-11}) at both $x = 3$ and $x = 7$.
- Symmetry: the answers are symmetric in a and b, as they must be, since the path is a parabola symmetric about $x = \frac{1}{2}(a+b)$.
- Units: ab/h and $(a+b)^2h/(ab)$ are lengths, matching $2V^2/g$; and $\tan\alpha$ is dimensionless.
- At $h = ab/(a+b)$ we get $\alpha = 45°$ and $2V^2/g = 2(a+b)$; for $a = 3$, $b = 7$ this is 20 m, slightly less than the 20.30 m above, consistent with it being the minimum.

Common slips

- Writing two separate equations for the walls and solving them simultaneously by brute force: it works, but the sum and product of roots does it in two lines.

- Forgetting that $\sec^2\alpha = 1 + \tan^2\alpha$ when converting k back into V.
- Quoting the sign of δV without factorising the derivative of $2V^2/g$: the clean condition comes from $h^2 > a^2b^2/(a+b)^2$ with $h > 0$.

Beyond STEP

The second part is a small sensitivity analysis: we linearised the map $h \mapsto (\alpha, V)$ around a solution and read off how the outputs respond. Linearisation, and the related idea of envelopes of families of trajectories (the "parabola of safety" bounding everything reachable at a given speed), are standard in first-year mechanics and calculus courses. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and these approximation techniques with care.

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